The Science Genome: A Framework for Semantic Scientific Analysis

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Abstract

This paper presents a standalone framework for analyzing how scientific ideas are inherited, transformed, and recombined across publication lineages. Publications are represented in a shared semantic space and decomposed into inherited content explained by cited predecessors plus residual contribution not explained by those predecessors. The framework combines constrained inheritance inference on a citation graph, enabling interpretable edge-level and node-level analysis of knowledge flow. The result is specification for content-aware scientometric analysis that supports lineage tracing, novelty assessment, and reproducible research interpretation.

1 Introduction

Scientific progress emerges through reuse, adaptation, and recombination of prior ideas. Citation networks record part of this process, but citation counts and centrality metrics alone do not reveal how intellectual content flows, or how contributions from predecessors impact those in newer papers.

This work formalises a content-aware framework for that distinction. Each paper is represented in a shared semantic space and approximated as a constrained mixture of cited-parent embeddings plus a residual term. The mixture weights encode lineage-specific inheritance, while the residual captures contribution not attributable to cited predecessors.

1.1 Problem focus

The framework is designed to support analysis of:

- estimating parent-specific inheritance strength for each citation edge,
- quantifying paper-level residual contribution not explained by cited parents,
- mapping branch-level continuity and separation from aggregated edge and residual signals,
- and measuring cross-field transfer through inheritance mass that crosses discipline labels.

1.2 Research stance

The objective is analytical interpretation, not causal proof of influence. Inheritance and contribution factors are evidential signals that require uncertainty-aware reading and expert contextualization.

1.3 Contributions of this paper

This paper delivers a final, standalone specification by providing:

- a formal decomposition model of parent-mixture inheritance and residual contribution,
- a constrained estimation procedure with uncertainty reporting for stable edge-level attribution,
- and a verification–validation protocol that tests specification compliance and interpretive utility.

2 Related Work

Research on scientific-publication analysis can be grouped into relational bibliometrics, semantic modeling, and hybrid science-of-science methods. Our framework builds on all three while targeting a specific gap: estimating inherited versus novel semantic content at the paper level.

2.1 Relational bibliometrics and citation-network analysis

Classical scientometrics treats citations as links in a graph and studies structure using count-based and network-based statistics. Citation totals, impact factor variants, and author-level indices provide simple, scalable proxies for influence, but they are coarse and field-sensitive. Network methods such as PageRank-style centrality, co-citation, bibliographic coupling, and community detection better capture structural position and diffusion pathways. Temporal extensions model growth, obsolescence, and burst dynamics across time slices.

These methods are valuable for mapping influence, yet they generally assume that each citation edge has comparable semantic meaning. In practice, citations can indicate foundational dependence, contrast, perfunctory acknowledgment, or methodological reuse. Without content, relational metrics cannot reliably distinguish these roles.

2.2 Semantic representations of scientific text

Semantic scientometrics introduces textual representations to recover meaning beyond graph topology. Early approaches relied on bag-of-words vectors and topic models; more recent work uses contextual embeddings from transformer models trained on scholarly corpora. Such representations support document similarity, clustering, trend analysis, entity linking, and literature retrieval.

A related line studies citation-context semantics by analyzing the sentence or paragraph surrounding citation mentions. This improves understanding of citation intent and rhetorical function. However, most semantic approaches still focus on similarity or classification and do not explicitly decompose a paper into inherited versus novel components.

2.3 Hybrid text–graph approaches

Hybrid methods combine citation structure with text features through graph neural models, heterogeneous networks, or joint embedding objectives. These systems improve tasks such as recommendation, field classification, and influence prediction by leveraging both topology and content. They demonstrate that semantic and relational signals are complementary.

Yet, even in hybrid settings, the output is often a latent score optimized for prediction tasks rather than an interpretable decomposition aligned with scientific contribution. This limits their use for transparent evaluation and policy-facing analysis.

2.4 Paradigm shifts and novelty detection

Work inspired by Kuhnian dynamics attempts to detect scientific revolutions through topic transitions, citation reconfiguration, and disruption indices. Novelty measures often rely on atypical combinations of references, semantic distance from prior work, or divergence from field norms. These methods offer useful indicators of change, but many are sensitive to parameterization and do not directly quantify what fraction of a paper is inherited from specific predecessors.

2.5 Gap addressed by this study

Across these literatures, no widely adopted framework simultaneously provides: (i) edge-level inheritance strength among cited parents, (ii) node-level residual novelty for each paper, and (iii) integration of both signals into a lineage-oriented citation DAG. The pseudo-genetic framework proposed here is designed to fill that gap with an interpretable, reproducible pipeline grounded in modern scientific-text embeddings.

3 Method

The method has four stages: (1) construct a semantically embedded paper corpus, (2) build a citation-based DAG, (3) estimate inheritance weights from cited parents to each focal paper, and (4) derive contribution and lineage statistics from the fitted decomposition.

3.1 Construction of science space

Let each paper p_i be represented by title and abstract text. We encode this text using an appropriate transformer (e.g. SciBERT) to obtain an embedding $\mathbf{e}_i \in \mathbb{R}^d$. Embeddings are ℓ_2-normalized to stabilize cosine geometry and reduce scale effects between documents of different length (large documents having large magnitudes and seen as "more simular" - this also means $\|p_i\| = 1$).

To improve robustness, we apply light preprocessing (Unicode normalization, citation-marker removal, and deduplication by identifier). We then compute corpus-level diagnostics (norm distribution, nearest-neighbor density, and year-stratified centroid drift) to verify embedding consistency before downstream estimation.

3.2 Construction of citation DAG

For each citation ($p_j \in \mathcal{P}(i)$) of paper p_i can be connected with a directed edge ($j \to i$) to build up a *Directed Acylic Graph* (DAG) with any circular citations are resolved by removing edges where publication year of p_i exceeds that of p_j.

3.3 Inheritance inference

We seek to decompose the semantic composition of p_i into:

1. the semantic contribution of each cited parent j on i (ϕ_{ij});

2. the magnitude of paper i's novel semantic content ($\|r_i\|$);

3. the semantic direction of that novel content ($\bar{r}_i$).

In a way that is rigorous and grounded in vector reconstruction and projection methods.

Paper i is decomposed into a reconstructed inherited component and residual component:

$$p_i = \sum_{p_j \in \mathcal{P}(i)} w^*_{ij} p_j + r_i$$

where w^*_j are constrained non-negitive reconstruction coefficients and r_i is the residual semantic component.

The coefficients w^*_{ij} are obtained via constrained least-squares reconstruction:

$$w^*_i = \arg\min_{w_{ij} \geq 0} \left\| p_i - \sum_{p_j \in \mathcal{P}(i)} w_{ij} p_j \right\|^2$$

subject to:

$$\sum_{p_j \in \mathcal{P}(i)} w_{ij} \leq 1$$

Novel Semantic Contribution

The semantic direction of th enovel contribution is represnted by the normalised residual direction $\bar{r}_i$, with the magnitude normalised to get the contribution magnitude:

$$\nu^*_i = \frac{\|r_i\|}{\sum_{p_j \in \mathcal{P}(i)} w^*_{ij} + \|r_i\|} \tag{1}$$

Parent Contribution

The raw reconstruction coefficients are not necessarily stable indicators of the contribution of paper j to paper i, as semantically similar papers may be interchangeable within the reconstruction. Parent contribution is therefore treated as an attribution problem rather than a purely geometric decomposition problem.

Parent contribution scores are computed using Shapley-value attribution, which measures the average marginal reconstruction contribution of each parent paper across all parent subsets.

In Shapley attribution, the value function defines the total utility produced by a subset of contributors. In this work, utility is defined as semantic reconstruction quality.

The Shapley attribution of parent paper j to target paper i is therefore given by:

$$\phi_{ij} = \sum_{S \subseteq \mathcal{P}(i) \setminus \{j\}} \frac{|S|!(|\mathcal{P}(i)| - |S| - 1)!}{|\mathcal{P}(i)|!} \Delta_i(j, s) \tag{2}$$

$$\Delta_i(j, s) = V_i(S \cup \{j\}) - V_i(S) \quad (3)$$

where $V_i(S)$ measures how well subset S reconstructs target paper i, and is defined as:

$$V_i(S) = 1 - \min_{\{w_{ij} \geq 0\}} \left\| p_i - \sum_{p_j \in S} w_{ij} p_j \right\|^2 \quad (4)$$

3.4 Handling non-orthogonal parents

Cited parents are typically correlated. To avoid unstable coefficient allocation, we combine three controls: (i) ridge regularization ($\lambda > 0$), (ii) optional parent-pruning by semantic redundancy threshold, and (iii) bootstrap re-estimation over parent subsets to compute confidence intervals for w_{ij}. We report both point estimates and uncertainty to distinguish robust inheritance edges from ill-conditioned assignments.

3.5 Derived metrics and analyses

From fitted weights we compute:

- **Edge inheritance strength** w_{ij} for lineage tracing.
- **Paper contribution and contribution magnitude** $\mathbf{r}_i$, and c_i or $\|\mathbf{r}_i\|_2$.
- **Lineage concentration** (entropy of $\mathbf{w}_i$), indicating whether inheritance is diffuse or dominated by a few parents.
- **Cross-field transfer** by aggregating inheritance mass across discipline labels.

These quantities support both ranking-style comparisons and structural analyses of knowledge evolution in the citation DAG.

4 Evaluation Protocol

4.1 Experimental objective

The experiments are designed to test the framework claims directly: that citation-conditioned semantic decomposition yields interpretable inheritance edges, meaningful residual contribution signals, and robust conclusions under perturbation.

4.2 Claim-driven evaluation skeleton

1. **Claim A: semantic reconstruction is informative rather than trivial.**
   - *Evidence target*: child embeddings are reconstructed by cited-parent mixtures with materially better fidelity than naive baselines.
   - *Suggested tests*: reconstruction error against (i) uniform parent weights, (ii) random cited-parent subsets, and (iii) popularity-weighted mixtures.
   - *Suggested metrics*: mean squared reconstruction error, cosine reconstruction score, and rank stability of high-contribution papers under metric choice.
2. **Claim B: inferred inheritance edges are numerically stable and interpretable.**
   - *Evidence target*: high-weight edges persist under parent-set perturbation and do not collapse under mild regularization changes.
   - *Suggested tests*: bootstrap re-estimation over parent subsets; sensitivity sweep over λ.
   - *Suggested metrics*: edge-weight confidence intervals, top-k edge overlap across runs, and variance of node-level lineage concentration.
3. **Claim C: residual contribution captures signal distinct from citation popularity.**
   - *Evidence target*: contribution ranking is not reducible to citation count, venue prestige, or recency.
   - *Suggested tests*: correlation and partial-correlation analyses against popularity-oriented indicators; matched-group comparisons.
   - *Suggested metrics*: Spearman/Pearson correlation, explained variance gap versus baselines, and divergence profiles of top-ranked papers.

4. **Claim D: representation quality supports downstream inference.**
    - *Evidence target*: science-space structure is coherent enough for inheritance decomposition to be meaningful.
    - *Suggested tests*: nearest-neighbor density summaries and domain-tagged low-dimensional maps for semantic clustering.
    - *Suggested metrics*: density-distribution summaries, within-domain versus cross-domain neighborhood similarity, and cluster-separation diagnostics.

4.3 Minimum evidence package for the draft

- Reconstruction-fidelity table with baseline comparisons.
- Inheritance-stability table (bootstrap confidence intervals and overlap statistics).
- Contribution-versus-popularity divergence analysis.
- Domain-tagged semantic map and nearest-neighbor density summary.

4.4 Failure and limitation disclosure template

- Missing or noisy citation edges that alter parent sets.
- Embedding degradation for domain-specific language and terminology drift.
- Collinearity-driven instability in weak inheritance edges.
- Interpretation risk where residual signal may conflate novelty and representation artifacts.

5 Discussion

The pseudo-genetic formulation reframes evaluation from visibility-only indicators toward compositional analysis of knowledge flow. By separating inherited semantic mass from residual contribution, the framework provides interpretable evidence at both citation-edge and paper levels.

5.1 How to read contribution and inheritance

A high contribution factor indicates lower explainability by cited parents under the selected representation; it is not a direct claim of quality or impact. A high inheritance factor indicates stronger semantic dependence on predecessors; it is not a deficit signal. Synthesis papers, benchmarks, and consolidation work can have high scientific value despite high inherited mass.

5.2 Analytical uses

The framework supports:

- lineage tracing for hypothesis-driven analysis,
- comparison of inheritance concentration across branches,
- detection of potential cross-domain transfer channels,
- and structured review of novelty claims against citation-supported dependence.

5.3 Interpretive boundaries

Outputs should not be used as deterministic judgments of researcher merit and should not be treated as causal evidence of intellectual influence without corroborating analysis. Strong claims should rely on convergence between quantitative outputs, uncertainty summaries, and expert qualitative inspection.

6 Limitations

6.1 Core limitations

The framework has structural limits that must be accounted for in interpretation:

- Embeddings compress complex text and may miss methodological detail.
- Citation graphs are incomplete proxies for true intellectual dependence.

- Correlated parent sets create attribution non-uniqueness without regularization.
- Residual contribution may reflect novelty, citation omission, or representation error.

6.2 Reporting requirements

Responsible use requires explicit reporting of:

- confidence or stability summaries for inheritance estimates,
- preprocessing and parameterization choices,
- distinction between descriptive evidence and causal interpretation,
- and sensitivity of conclusions to weak or unstable edges.

These requirements are part of the framework definition and are necessary for publishable, reproducible analysis.

6.3 Open decisions and unresolved directions

- Whether to treat nearest-neighbor density primarily as an evidence metric or as a hard quality gate in the execution pipeline.
- How to interpret year-stratified centroid drift: temporal language shift, methodological trend shift, or mixed effects.
- Whether temporal drift should be discounted from author-level novelty attribution, and if so, by what correction scheme.
- Which uncertainty threshold should separate robust inheritance edges from exploratory-only edges in publication-facing analyses.

7 Conclusion

This paper presents a final standalone framework for content-aware analysis of scientific idea flow. The framework combines semantic decomposition and citation lineage structure to estimate inherited dependence and residual contribution in an interpretable form.

By coupling the model with explicit verification, validation, uncertainty reporting, and interpretation guardrails, the artifact supports rigorous and reproducible scientometric inquiry rather than opaque scoring.

The resulting specification is implementation-agnostic, self-contained, and suitable as a publishable basis for studying how scientific ideas accumulate, recombine, and diverge.

References

TEMP: Todo

- Go through current paper, making sure everything I currently want in it is in it, and that it is written from the perspective of the paper, not from the perspective of my task (I am sus about the V&V parts being in there)
- Make implementation lean and suplimentary to paper. Run investigations for the testing and validation
- Update the paper with what was seen in the testing and validation, any additional maths, etc etc
- Add diagrams to help explian concepts and visualise results (the topic map, inherritence etc) - make sure to only add things that add, keep them minimal and professional looking, and remember the intent - it isn't just to make it look fancy, it is to supliment the reading and make the concepts more intuitive.

- Add references where needed - if I make a claim that isn't mine, or reference a prior idea, add a reference to it
- Review for next steps